

Pranjal Tiwari

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Education

Indian Institute of Science Education and Research Mohali
BS-MS (Physics Major)
GPA : 8.50/10

2022- present

Talks

- I gave a 15 minute talk based on the paper "Black Holes and the Second Law" by J. Bekenstein (1972)([Slides](#))
 - Presented our work "Obviating PBH Overproduction for SIGWs generated by PTAs in loop-corrected EFT of bounce" at the Cosmology and Gravitational Waves Journal Club, Swansea University([Slides](#))
 - Delivered an introductory talk on cosmology and inflation for undergraduate students at the Astronomy Club, IISER Mohali ([Slides](#),[Video](#))
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Masters Thesis

Quantum Field Theory in De Sitter

Duration: June 2026- Present

Supervisor: [Dr. Arnab Priya Saha](#)

Description: Working on representation theory of particles in de Sitter space.

Summer Projects

Regularization of Quantum Field Theories

Duration: May 2025- Aug 2025

Supervisor: [Prof. Ambresh Shivaji](#)

Description: Worked on one-loop diagram calculations and studied multiple regularization methods.

Inflationary Cosmology

Duration: May 2024- Aug 2024

Supervisor: [Prof. Sayantan Choudhury](#)

Description: Studied the formation of primordial black holes and scalar-induced gravitational waves from early-universe density perturbations. Also worked on an inflationary model addressing PBH overproduction, leading to a publication([link](#))

Academic achievements

- GATE Physics 2026: All India Rank- 166
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Relevant Courses

Physics (Graduate Courses)

- Quantum Field Theory I
- Quantum Field Theory and the Standard Model
- Quantum Fields in Curved Spacetime
- General Theory of Relativity
- Statistical Field Theory
- Quantum Phases of matter and phase transitions
- Quantum Information Theory
- Astrophysics of Stars and Galaxies
- Solid State Physics
- Nuclear and Particle Physics

Mathematics(Undergraduate Level Courses)

- Introduction to Differential Geometry
 - Group Theory and Linear Algebra
 - Real Analysis
 - Probability and Statistics
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Publications

- S. Choudhury, K. Dey, S. Ganguly, A. Karde, S. K. Singh, and P. Tiwari, "Negative non-Gaussianity as a salvager for PBHs with PTAs in bounce." [Eur.Phys.J.C 85 \(2025\) 4, 472](#), [arXiv:2409.18983](#)
 - S. Choudhury, S. Ganguly, S. Panda, S. SenGupta, and P. Tiwari, "Obviating PBH overproduction for SIGWs generated by Pulsar Timing Arrays in loop corrected EFT of bounce." [JCAP 09\(2024\) 013](#), [arXiv:2407.18976](#)
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Skills

- Python
 - Mathematica
 - Linux
 - Bash Scripting
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